

Tailoring Machine Learning Weather Predictions for Local Impacts

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What's Been Happening in Weather Forecasting?

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- Beginning in 2022, several machine learning (ML) based weather models have been created that have rivaled or exceeded the accuracy of traditional models.
- Critically, these models can be run on a personal laptop.

Increased Accessibility of Weather Predictions Opens New Opportunities

- Individual countries and smaller organizations can now run their own weather models instead of relying on large government agencies.
- Global models are more accurate in some areas than others.
- Different groups prioritize accuracy for different weather variables.

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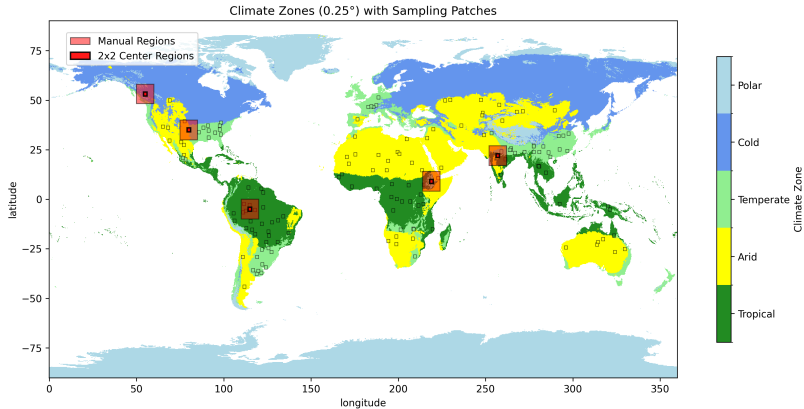
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- **The Rub:** All global weather models (NWP or ML) are trained to optimize overall global accuracy.
 - A temperature error in Greenland is treated the same as a temperature error in Chicago!
 - The ML models might be cheap to train, but they are extremely expensive to train!
- **Potential Solution:** Post-process global model outputs with a small, localized model to improve accuracy for specific regions and weather variables of interest.

How to Post-Process Weather Model Outputs? What Outputs Matter Most?

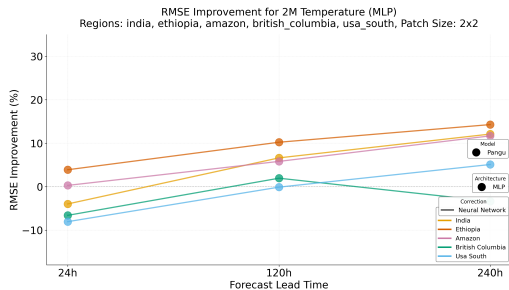
- Just as ML tools can be used to post-process models just as they have been used to build them.
 - **Contribution:** Demonstrate that a simple low-compute neural network model can meaningfully improve the accuracy of both traditional and ML weather models. Additionally, show where these simple models perform well, and where they struggle.
 - Additionally, show that post-processing using alternative loss functions can lead to greater improvements for specific weather variables of interest.

Start by Manually Looking at Five Regions of Interest

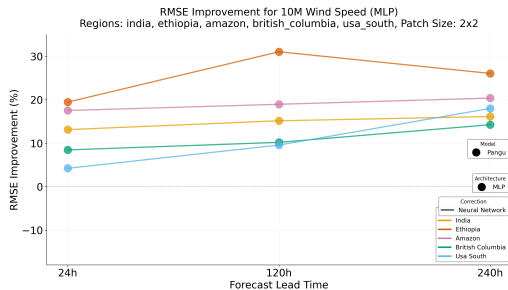
- Manually choose a few regions with a variety of climates and land terrains.
- Sample from different climate zones.



See Greater Improvements for Longer Lead Times and for Wind Speed

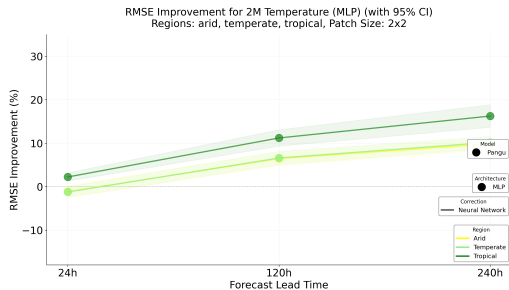


(a) 2m Temperature: Pangu (ML Model)

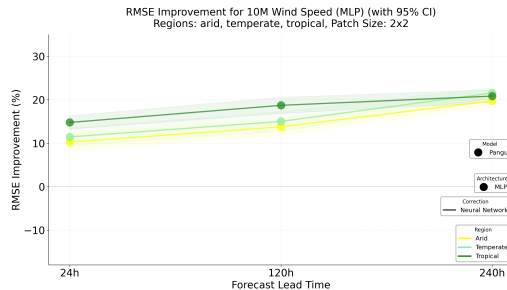


(b) Wind Speed: Pangu (ML Model)

Climate Zones



(a) 2m Temperature: Pangu



(b) Wind Speed: Pangu